

We want to estimate the driving power W necessary to move at a peak velocity $V = 10 \text{ m/s}$ a marine freighter of length $l = 100 \text{ m}$ and total mass at full load $m = 1000 \text{ Tm}$. For that purpose, a small model of length $l_e = 1 \text{ m}$, geometrically similar to the real ship, is to be tested on a hydraulic channel. To guide the design of the experiment, follow these steps:

1. Use the dimensional analysis to reduce the parametric dependence of the required power W .
2. Write the three equations that need to be satisfied to guarantee physical similarity between the experiment and the full-scale prototype. Verify that it is not possible to find the values of the mass m_e and velocity V_e of the experiment that satisfy simultaneously all three equations, so that complete physical similarity cannot be attained.
3. Assuming that the effect of viscosity on the fluid motion is negligible, determine the values of m_e and V_e . Check that the corresponding values of the Reynolds number for both the real prototype and the experiment are indeed sufficiently large for the assumption of negligible viscosity to be a reasonable approximation.
4. If the experimental measurement of the power obtained with the experimental model is $W_e = 10 \text{ W}$, obtain the value of the driving power W needed to operate the real prototype at the required conditions.

① $W = W(V, g, m, l, \rho, \mu) \rightarrow \frac{W}{\rho V^3 l^2} = f\left(\frac{g}{V^2/l}, \frac{m}{\rho l^3}, \frac{\mu}{\rho V l}\right) = f\left(\frac{V^2}{g l}, \frac{m}{\rho l^3}, \frac{\rho V l}{\mu}\right)$

② $\frac{V^2}{g l} = \frac{V_e^2}{g l_e} \rightarrow V_e = \left(\frac{l_e}{l}\right)^{1/2} V$
 $\frac{V l}{\rho} = \frac{V_e l_e}{\rho} \rightarrow V_e = \frac{l_e}{l} V$
 $\frac{m}{\rho l^3} = \frac{m_e}{\rho l_e^3} \rightarrow m_e = \left(\frac{l_e}{l}\right)^3 m$

③ $\left[V_e = \left(\frac{l_e}{l}\right)^{1/2} V = 1 \text{ m/s}\right], \left[m_e = \left(\frac{l_e}{l}\right)^3 m = 1 \text{ kg}\right]$

④ $\left[W = \left(\frac{V}{V_e}\right)^3 \left(\frac{l}{l_e}\right)^2 W_e = 100 \text{ MW}\right]$

FRUENE NUMBER
 REYNOLDS NUMBER
 PROPORTIONAL TO THE SUBMERGED-TO-TOTAL SHIP VOLUME RATIO

$Re = \frac{V l}{\nu} \approx 10^9 \gg 1$
 $Re_e = \frac{V_e l_e}{\nu} = 10^6 \gg 1$